

MMMB215 – Project 2 Report

4 members worked on this project

Statement of Contribution

(tick one)

☐ We confirm and agree that all the team members have made **equal contributions** to the project.

☒ **We confirm and agree that our contributions to the project vary in their extent and significance.
The contribution of each team member to the specific tasks is as noted below:**

Introduction: Team member X

Background Research: Team member X

Task 1: Team member Y

Task 2: Team member Z

Task 3:

- Fan shaft design: Team member Z
- Analytical calculations: Ghad Saqqa
- FEA: Team member X
- Analytical and FEA comparison: Team member X

Task 4: Team member X

Task 5:

- Design brackets and use cap screws:
- Select arrange and size of screws/bolts: Ghad Saqqa
- Calculate required screws/bolts numbers and FOS: Ghad Saqqa

Task 6: Team member Z

Formatting: Team member X

Note that if this form is not provided in your report, all the team members are considered contributing equally.

Introduction

Project 2 looks at designing the mechanical system of an axial-flow tunnel ventilation fan. Transport tunnels use these fans to maintain clean and safe air by removing gases, dust, and heat during normal operation. In emergencies such as fires, they also help control smoke and maintain clear evacuation paths.

This project prioritises mechanical aspect of how power is transmitted from the motor to the fan and how the system is supported. The fan needs to be able to run for at least five years and give off 45 kW of shaft power at 500 rpm.

This involves selecting a suitable motor, designing a belt-drive system, creating fan shaft and bearing setup, and developing a ceiling mounting to hold the entire assembly. Each task aims to produce a design that is safe, durable, and practical to manufacture.

Background Research

Design and Specification Limitations

The first step was identifying the specification requirements and the design restrictions. According to the specifications, the system must have a minimum continuous service life of five years, a fan-shaft power of 45 kW, and a rotational speed of 500 rpm. The design must be accomplished in accordance with the constraints, which include using an AC induction motor-powered belt-drive transmission. Recognising these limits influenced the design process and directed the investigation of each key element.

Axial-Flow Fan Fundamentals

An axial-flow fan moves air along the shaft of rotation, which allows for a lot of airflow at a low pressure. This kind of fan is great for ventilating tunnels because it keeps a steady flow of air along its length and is small. Instead of directly connecting the motor to the fan, a belt system connects them. This cuts down on vibration, makes maintenance easier, and makes it easier to line up the fan while it's running.

Motor Selection and Belt Drive Design

The motor needs to be able to run at 500 rpm all the time and provide enough power and torque to meet the 45 kW requirement. The choice depends on the rated power, efficiency, frame size, and type of mounting. You can use manufacturer catalogues to make sure that the parts are compatible with the rest of the system and meet the specifications.

The belt drive moves torque from the motor to the fan while slowing it down. Using data from the manufacturer, you can choose the pulley sizes, belt types, and number of belts to make sure the drive can handle full-load conditions safely. For your machine to work well and last a long time, the belt must be properly aligned and tensioned.

Shaft and Bearing Design Considerations

The fan shaft supports the pulley, hub, and blades while transmitting torque through the system. The belt and fan thrust put a mix of twisting, bending, and axial loads on it, so it needs to be strong and stiff enough to handle these without bending.

Bearings are chosen because they can carry both radial and axial loads and stay in line. Keeping the right amount of space between them helps keep the shaft from bending and vibrating. Also, fillets, shoulders, and keyways are important because if they aren't designed correctly, they can cause stress concentrations. The overall goal of the shaft layout is to find a balance between strength, reliability, and ease of assembly.

Ceiling Mounting and Fasteners

The entire weight of the motor, fan, and transmission system is supported by the ceiling mounting structure. For it to withstand both static and dynamic loads, it must have sufficient strength and rigidity. Cap screws are sized with appropriate safety factors based on tensile and shear limits. The bracket geometry is made for easy installation, safe anchoring, and vibration stability.

Analytical Analysis:

The analytical approach determines the stresses acting on the shafts, including torsional shear stress and bending stress, at their most critical sections using fundamental equations of mechanics. This is then followed by combining them using the Goodman criteria to find the factor of safety under fully reversed loading conditions, the Von Mises criterion is then used to find the equivalent stress. This ensures that the shafts can withstand the maximum permitted torque and speed defined by the project specifications.

Numerical Analysis:

The numerical approach involves conducting a Finite Element Analysis (FEA), a computer-based simulation technique that simulates the behaviour of an actual object under a range of physical conditions, including vibration, heat, and stress. Using software such as ANSYS which simulates real-world loading on the shafts, accounts for the detailed shaft geometry, including shoulders, keyways, and bearing locations, and allows stress concentrations to be accurately captured. The results of the simulation are then compared to the results of the analytical method to verify the accuracy of the calculations and reveal any differences between the theoretical and the simulated approach.

Task 1: Motor Selection

Design requirements for the motor are that it must transmit at least 45 kW of power and 500 rpm in the output shaft. It must also be a generic commercial AC induction motor. It must also have a service life of 5 years while in constant operation. Outside of these direct requirements we are also aware that it is going to be used to power an Axial Flow Tunnel Ventilation Fan (Fan), which are usually mounted to the ceiling, as such we want to reduce the weight of each component, including the motor as much as possible. Browsing TECO results in 9 options for 45 kW motors, all of which are 3 phase AC induction.

Table 1: Motor Results from TECO Motor Selector

Motor Name	Notes
TECO MAXe3 MINING High Efficiency	High-efficiency cast iron motor for industrial fans. [1]
TECO MAXe3 Premium Severe Duty	Excels in extreme weather conditions without failing, dusty, wet conditions. [1]
TECO IE3 Flameproof (Exd)	Used for environments where exposed to explosive gas. Used in safety critical situations. [2]
TECO IE2 Flameproof (Exd/Exde)	Flameproof option at lower efficiency. [3]
TECO MAX-Ex (Exe,ExnA,ExtD)	Hazardous area motors with options for dust, water and flame proofing. [4]
TECO Open Drip Proof (ODP)	Typically used in clean plants when not exposed to adverse conditions. [5]
TECO AEBH Vertical Hollow Shaft Motor	Used in vertical shafts, vertical pumps etc. Not common for horizontal fans. [6]
TECO Te3 Cast Iron 3 Phase	General purpose cast-iron, standard industrial motor for direct-drive axial fans. [7]

Based on the brochures available, it makes the most sense to use a TECO Te3 Cast Iron 3 Phase Motor (Te3). The severe duty and flameproof motors are typically used in extreme weather conditions, which, unless specified otherwise is not the typical application of the Fan. The ODP motor is typically only used in very clean environments and would not be able to survive typical tunnel conditions. A Vertical Shaft Motor does not fit standard fan designs and would not be recommended for typical use in an axial flow fan. Te3 provides a sufficient power supply at a typically relatively low weight when compared to its heavy-duty counterparts. If this were not available, the next best recommendation would be the TECO MAXe3 MINING High Efficiency Motor, due to its frequent application in mining and adequate protection at typically low weight and cost. Typical weight values are difficult to determine without inspecting site and determining application; however, the selected motor is typically lighter and cheaper than the alternatives.

Task 2: Belt Drive Design

There are two choices that can be made first in belt designs, that being either a flat belt or a v belt. Flat belts have a rectangular cross section whose width is greater than its thickness and only the bottom surface contacts the pulley, requiring a great deal of friction to prevent slippage [8]. As such slip and creep can frequently occur. With that said, flat belts are smooth, can absorb shock and operate with flexibility and high efficiency at speed [9]. Comparatively V belts have a trapezoidal shape and make contact at both sides with the pulley, and as it is a positive drive it operates with no slip as well as greater friction, the V belt can transmit greater power than the flat belt. It also operates much better at shorter distances because of the flat belts tendency to slip due to the tight side at short distances. Since our application is for that of a fan, the belt length will not exceed 1m much at all. Flat belts tend to have an efficiency around 99% with V belts only slightly less efficient ~96-97%. Both can be arranged in various arrangements with idler pulleys, compound belts, crossed belts etc. [10]

For these reasons we elected to choose a v belt design with a metre separation between pulleys. Browsing SKF for V belts transmitting 45kW of power and driving at 1000rpm being reduced to at least 500rpm, we were recommended the SPB XP2 belt with the following data:

Table 2: Calculation results from SKF Belt Calculator

Input Data	
Rated power [kW]	45
Rated torque [Nm]	429.72
Requested service factor	1.4
Driver speed [rpm]	1000
Driven speed [rpm]	568.31
Overall width [mm]	101.00
Drive	
Overall length [mm]	1,325.31
Overall Height [mm]	407.00
Pulley center distance [mm]	1,006.31
Span length [mm]	1,002.46
Actual driven speed [r/min]	568.31
Belt	
Designation	PHG SPB3000XP2
Number of belts / grooves	5
Belt width [mm]	16.00
Actual service factor	1.42
Belt power rating [kW]	64.03
Belt mass [kg]	0.67
Driver Pulley	
Designation	PHP 5SPB224TB
Reference diameter [mm]	224.00
Driven Pulley	
Designation	PHP 5SPB400TB
Reference diameter [mm]	400.00
Hub Loads	
Static hub loads for NEW belt [N]	10,401.83
Static hub loads for RUN-IN belt [N]	6,934.55
Dynamic hub loads for NEW belt [N]	10,070.67
Dynamic hub loads for RUN-IN belt [N]	6,603.40
Tensioning	
Installation strand tension for NEW belt [N]	1,040.18
Installation strand tension for RUN-IN belt [N]	693.46
Requested service factor	6.76
Driver speed [r.min]	4.55
Driven speed [r.min]	33.91
Driven speed [r.min]	27.69

Table 3: Additional Belt Calculations from SKF Belt Calculator

Belt	
Wedge SPB XP	
Belt marking	SPB3000XP
No. of ribs	1
Pitch length (mm)	3000
Effective length (in)	118.1
w = Width (mm)	16.3
h = Height (mm)	13

Table 4: Additional Driver Pulley Calculations from SKF Belt Calculator

Driver Pulley	
Pitch diameter (mm)	224
Outside diameter (mm)	231
Pulley type	3
Bush no.	3020
Min. bore (mm)	25
Max. bore (mm)	75
F	101
E	-
G	181
K	25
L	51
M	25
H	-
Weight (lbs.)	13.7

Table 5: Additional Driven Pulley Calculations from SKF Belt Calculator

Driven Pulley	
Pitch diameter (mm)	224
Outside diameter (mm)	231
Pulley type	3
Bush no.	3020
Min. bore (mm)	25
Max. bore (mm)	75
F	101
E	-
G	181
K	25
L	51
M	25
H	-
Weight (lbs.)	13.7

Task 3: Fan Shaft Design

Manual Calculations

Using AS/NZS 3678 - 250 XLERPLATE® steel

Table 6: Shaft 1 and 2 Specifications

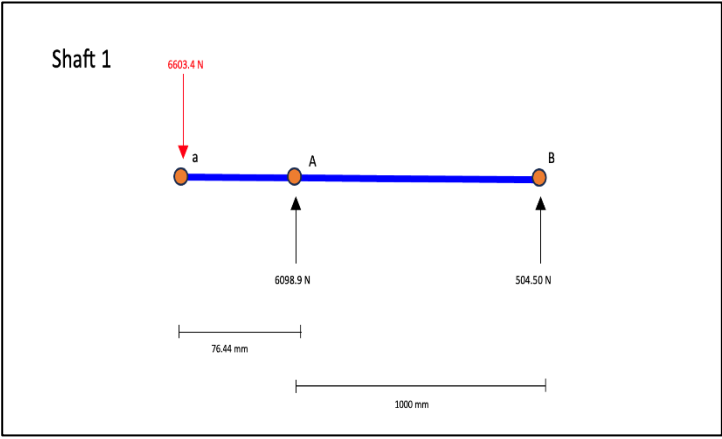
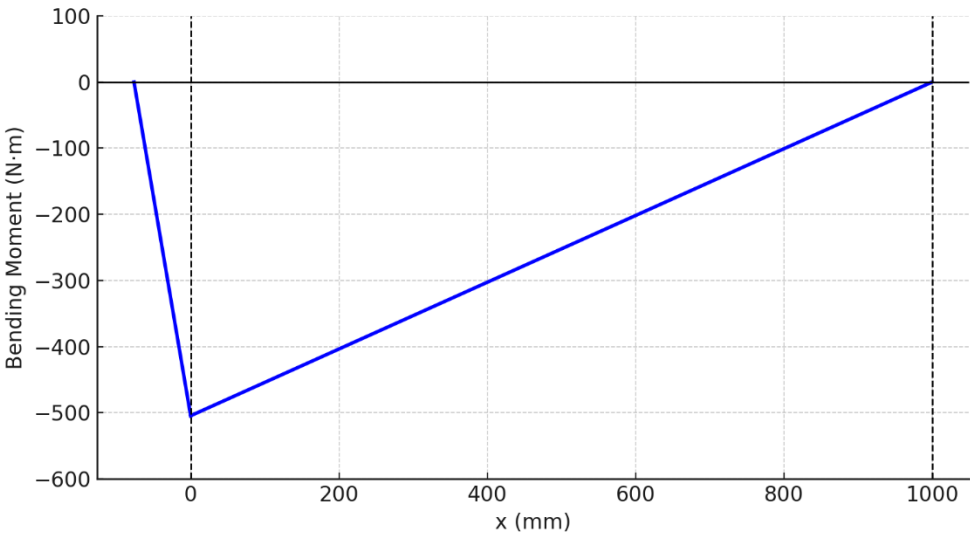
	Shaft 1 – Internal Shaft	Shaft 2 – Top shaft
Power	45 kW	45kW
Speed (rpm)	568.31 rpm (actual fan speed from SKF)	1000 rpm (motor speed from SKF)
Torque (Nm)	756.13 N.m	429.72 N.m
Load (N)	6603.40 N (dynamic hub load RUN-IN belt, SKF)	6603.40 N (equal and opposite belt load)
Geometry	Stepped solid shaft: Ø127 → Ø50 mm	Stepped solid shaft: Ø127 → Ø50 mm
Critical Sections	Small Ø50 mm end (shoulder transition under torsion + bending)	Small Ø50 mm outer (Ø25 mm inner) end under torsion and bending
Max Shear stress (MPa)	30.81 MPa	18.49 MPa
Von Mises Stress (MPa)	67.36 MPa	72.54 MPa
Static FOS	3.86	3.02
Fatigue FOS	1.77	1.23
Notes/Assuptions	$S_y = 260 \text{ MPa}$ $S_{UT} = 410 \text{ MPa}$ Bending was taken as a fully reversed loading: $\sigma_{m,b} = 0, \sigma_{a,b} = \sigma_b$ $\tau_a = 0, \tau_m = \tau$ $\sigma_{a,eq} = \sqrt{\sigma_{a,b}^2 + 3\tau_a^2} = \sigma_{a,b}$ $\sigma_{m,eq} = \sqrt{\sigma_{m,b}^2 + 3\tau_m^2} = \sqrt{3}\tau_m$ Endurance limit for steels $S'_e \approx 0.5S_{ut}$ as assumed common factors for unspecified steels. Common correction factors for surface, size, loading, reliability $k_a \times k_b \times k_c \times k_d \times k_e \approx 0.5$ Overhead load factor: $\gamma=2.5$ for ceiling-mounted assemblies (if applicable). The critical section is taken at the smallest shaft diameter, where combined torsional and bending stresses are highest. Torsion was the dominant component of the Von Mises stress, making shoulder transitions the most critical locations. Features such as fillets or keyways were not explicitly modelled due to the high safety factors achieved; therefore, the shaft is expected to remain structurally sound under the applied loads.	

Shaft 1 – Internal shaft

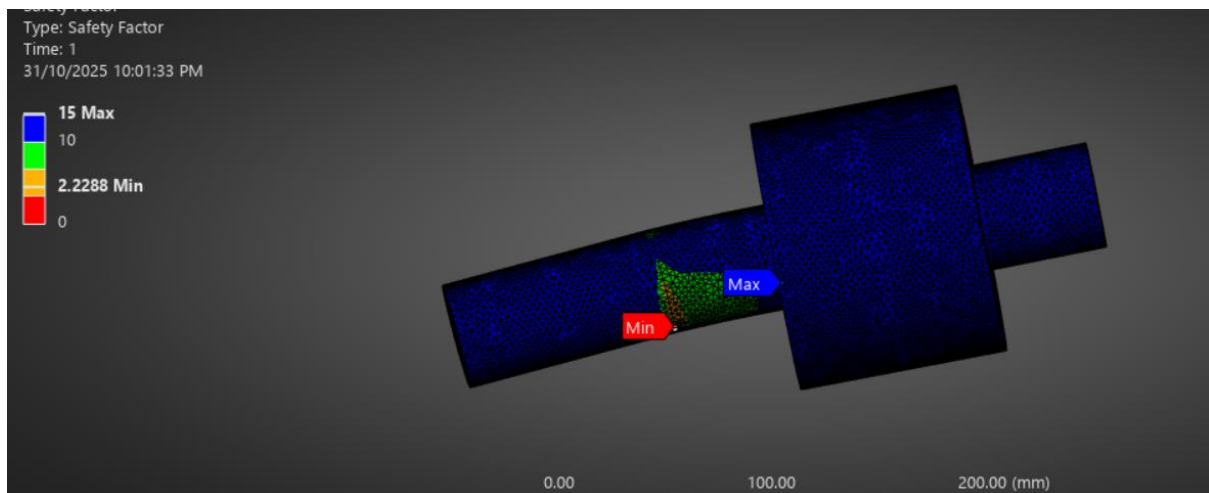
To determine the correct FBD, it is crucial to find the overhang and to determine where the load acts. To determine it, the model of the brush was crucial since a PHF TB 3020X is used for shaft 1; it can be found using the SKF catalogue and calculations, $L_{bush} = 50.8 \text{ mm}$, $t_{hub} = 51.0 \text{ mm}$ and a clearance of 0.1 mm was included in the initial design, which can be used to find the overhang as

$$a_{motor} = c + L_{bush} + \frac{t_{hub}}{2} = 0.1 + 50.8 + \frac{51.0}{2} = 76.4 \text{ mm}$$

Table 7: Shaft 1 analytical calculations

Reaction Equations	$\sum M_A = 0 = 6603.4(76.4) + R_{By}(1000) = 0, R_{By} = 504.50 \text{ N}$ $\sum F_y = 0 = -R_{Ay} + R_{By} - 6603.4 = 0, R_{Ay} = 6098.9 \text{ N}$
FBD	
Bending Moment Equations	$-a \leq x \leq 0, -76.4 \leq x \leq 0, M(x) = -F(x + a)$ $0 \leq x \leq L, 0 \leq x \leq 1000, M(x) = -Fa + \frac{Fa}{L}x$
Bending Moment Points	$M(-76.4) = 0$ $M(0) = -504.50$ $M(1000) = 0$
BMD	

Bending Stress	$\sigma_b = \frac{32M}{\pi d^3} = \frac{32 (504.5 \cdot 10^3)}{\pi \cdot 50^3} = 41.11 \text{ MPa}$
Shear Stress (Torsion)	$\frac{16T}{\pi d^3} = \frac{16 (756.13 \cdot 10^3)}{\pi \cdot 50^3} = 30.81 \text{ MPa}$
Von Mises Stress	$\sigma_{vm} = \sqrt{(\sigma_b^2 + 3\tau^2)}$ $\sigma_{vm} = \sqrt{(41.11^2 + 3 (30.81^2))} = 67.36 \text{ MPa}$
Static FOS (Yielding)	$n_y = \frac{\sigma_y}{\sigma_{vm}} = \frac{260}{67.36} = 3.86$
Modified Goodman (Fatigue)	$S'_e \approx 0.5 \cdot 410 = 205 \text{ MPa}$ $S_e = 0.5 \cdot 205 = 102.5 \text{ MPa}$ $\frac{\sigma_b}{S_e} + \frac{\sigma_{vm}}{S_{UTS}} = \frac{1}{FoS}, \quad \frac{41.11}{102.5} + \frac{67.36}{410} = 0.565$ $FoS = \frac{1}{0.565} = 1.77$



We obtained a FoS of 2.2288, which is very close to the manually calculated value of 1.77, indicating good accuracy.

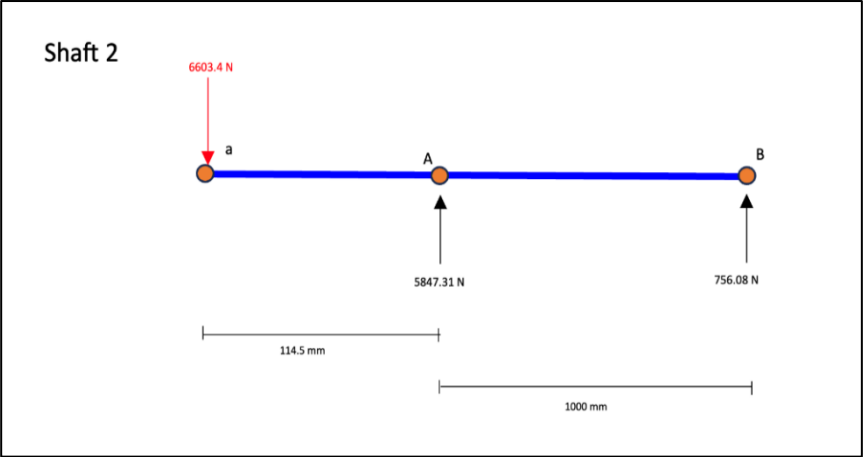
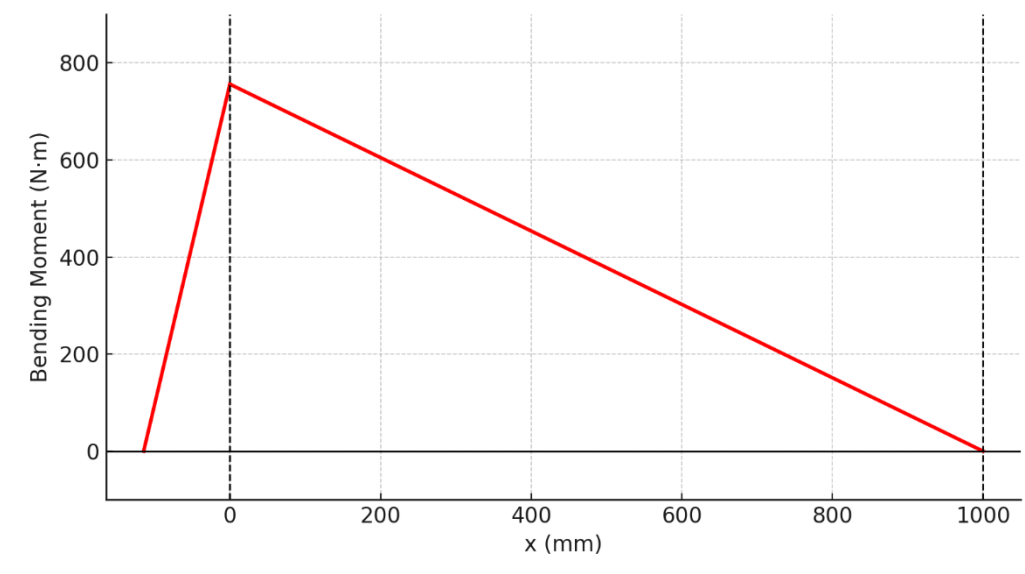
Shaft 2 – Top shaft

Similar to shaft 1, the same procedure is followed; however, this shaft used a PHF TB 3535X:

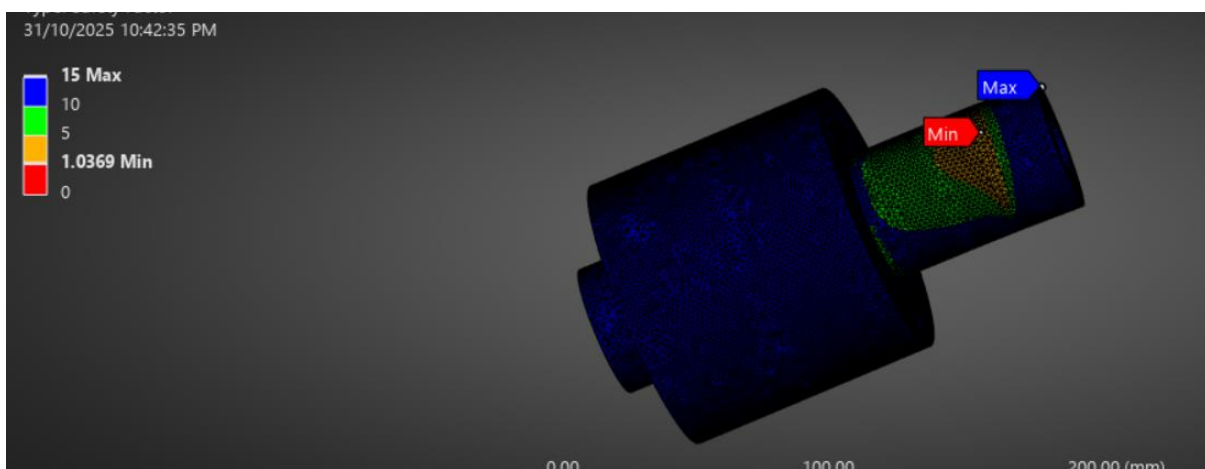
$L_{bush} = 88.90 \text{ mm}$, $t_{hub} = 51.0 \text{ mm}$ and clearance is 0.1 mm. Therefore, the overhang was found to be:

$$a_{motor} = c + L_{bush} + \frac{t_{hub}}{2} = 0.1 + 88.90 + \frac{51.0}{2} = 114.5 \text{ mm}$$

Table 8: Shaft 2 analytical calculations

Reaction Equations	$\sum M_A = 0 = 6603.4(114.5) + R_{By}(1000) = 0, R_{By} = 756.08 \text{ N}$ $\sum F_y = 0 = -R_{Ay} + R_{By} - 6603.4 = 0, R_{Ay} = 5847.31 \text{ N}$
FBD	
Bending Moment Equations	$-a \leq x \leq 0, -114.5 \leq x \leq 0, M(x) = F(x + a)$ $0 \leq x \leq L, 0 \leq x \leq 1000, M(x) = Fa + \frac{Fa}{L}x$
Bending Moment Points	$M(-114.5) = 0$ $M(0) = 756.09 \text{ N}$ $M(1000) = 0$
BMD	
Bending Stress	$\sigma_b = \frac{32MD_0}{\pi(D_o^3 - D_i^3)} = \frac{32(756.09 \cdot 10^3) \cdot 50}{\pi \cdot (50^4 - 24^4)} = 65.07 \text{ MPa}$
Shear Stress (Torsion)	$\frac{16TD_0}{\pi(D_o^4 - D_i^4)} = \frac{16(429.72 \cdot 10^3) \cdot 50}{\pi \cdot (50^4 - 24^4)} = 18.49 \text{ MPa}$
Von Mises Stress	$\sigma_{vm} = \sqrt{(\sigma_b^2 + 3\tau^2)}$

	$\sigma_{vm} = \sqrt{(65.07^2 + 3 (18.49^2))} = 72.54 \text{ MPa}$
Static FOS (Yielding)	$n_y = \frac{\sigma_y}{\sigma_{vm}} = \frac{260}{86.07} = 3.02$
Modified Goodman (Fatigue)	$S'_e \approx 0.5 \cdot 410 = 205 \text{ MPa}$ $S_e = 0.5 \cdot 205 = 102.5 \text{ MPa}$ $\frac{\sigma_b}{S_e} + \frac{\sigma_{vm}}{S_{UTS}} = \frac{1}{FoS}, \quad \frac{65.07}{102.5} + \frac{72.52}{410} = 0.812$ $FoS = \frac{1}{0.812} = 1.23$



We obtained a FoS of 1.0369, which is very close to the manually calculated value of 1.23, indicating good accuracy.

Task 4: Bearing Selection

4.1 Bearing Arrangement and Interface

In the final design, the fan system is vertically mounted, with the **motor positioned above and outside the casing**. The motor shaft carries the **driver pulley**, which transmits torque through a belt to the **driven pulley** mounted on the fan shaft inside the casing. The fan shaft is supported by two bearings located on either side of the casing wall — the upper bearing acts as the **locating bearing**, while the lower bearing functions as the **floating bearing**. This arrangement allows the belt load to be carried effectively by the upper support while permitting the shaft to expand freely during operation without introducing thermal stresses.

A **SKF 24122 CCK30/C3W33 spherical roller bearing** was chosen for the upper locating position. This bearing can accommodate heavy radial loads, moderate axial loads, and angular misalignment caused by belt tension or bracket deflection. The **SKF NU 2222 ECM cylindrical roller bearing** is used at the lower end. It provides high radial load capacity and allows free axial movement of the inner ring, making it ideal as a floating bearing. Together, they form a durable and well-balanced bearing arrangement suitable for vertical fan applications.

4.2 Bearing Loads and Calculations

The main radial load on the bearings comes from the belt tension acting on the driven pulley, estimated at **6.6 kN** during steady operation. A small additional axial load of about **0.33 kN** (5 % of the radial load) was included to account for the downward thrust of the fan blades.

For the spherical roller bearing, the equivalent dynamic load is given by:

$$P = F_r + Y_1 F_a$$

where $F_r = 6600\text{N}$, $F_a = 330\text{N}$, and $Y_1 = 1.8$.

$$P = 6600 + 1.8(330) = 7194\text{N}.$$

The minimum required dynamic rating for five years of continuous service ($L_{10h} = 43800\text{h}$ at $n = 500\text{rpm}$) is:

$$C = P \left(\frac{60nL_{10h}}{10^6} \right)^{1/3} \approx 78500\text{N}.$$

The selected SKF 24122 CCK30/C3W33 has a dynamic capacity of over **1 000 kN**, which provides an excellent margin of safety. Its static rating of approximately **27 000 kN** also far exceeds the maximum static equivalent load of **7 kN**, confirming that both load and fatigue requirements are easily met.

4.3 Fits, Mounting and Lubrication

The upper bearing is mounted with an **interference fit** on the shaft (H7/k6) and a **light press fit** in the housing (H8), secured axially by a shaft shoulder and locknut. The lower cylindrical bearing has a **slip fit** on the shaft to permit axial movement during expansion and is retained in position by a simple locating ring. Both housings include grease fittings and seals. The W33 lubrication groove on the spherical bearing allows for periodic relubrication, while the lower bearing is protected against dust and moisture by a sealed cover on the underside of the casing.

4.4 Service Life and Maintenance

Based on the calculated loads and operating speed, the expected bearing life far exceeds the minimum design requirement of five years of continuous operation. The bearings should be relubricated every **5 000 hours** or at least once per year, depending on operating conditions. Routine inspections for vibration, noise, and temperature rise should be conducted as part of the maintenance schedule. The open, externally mounted configuration simplifies access and replacement if needed.

4.5 Summary

The selected **SKF 24122 CCK30/C3W33 spherical roller bearing** (locating) and **SKF NU 2222 ECM cylindrical roller bearing** (floating) provide an efficient and reliable support system for the vertically mounted fan shaft. This arrangement supports combined radial and axial loads, compensates for misalignment, and accommodates thermal expansion. The configuration shown in the CAD model—where the motor and pulley assembly are mounted above the fan casing—accurately reflects the final design intent and ensures smooth, long-term operation with minimal maintenance.

Task 5: Ceiling Mounting Design

Since there is a wide variety of bolts, screws, and washers to pick from, the team conducted in-depth research in which it was concluded that using **ISO hex-flange cap screws** with Nyloc lock nuts would be the most appropriate when designing a ceiling bracket because their integrated flange distributes the load over a larger area, which increases the clamping force and prevents damage to the bracket's surface. This integrated washer design simplifies installation by eliminating the need for a separate washer, reducing installation time and cutting down on inventory. Additionally, the flange head gives a built-in bearing area, so on a flat steel bracket you don't need separate washers, less hardware, better load spread, less brinelling. A Nyloc (or prevailing torque) nut + medium thread locker handles vibration from the fan/motor. Material is A2/A4 stainless if corrosion is a concern; otherwise, property class 8.8 (or 10.9) for higher strength. "Partially threaded" bolts don't add value here (the joint is shear-dominated and the flange already provides bearing), and smaller machine screws are under-spec for a suspended 45 kW unit. This design assumes that the overall mass of the design = 500kg, and plate thickness is 8 mm.

Table 9: Bolt analysis and selection

Item	Formula/ Theory	Result
Assumptions	A2-70: $F_u = 700 \text{ MPa}$ $S_p = 450 \text{ MPa}$ 8.8: $F_u = 800 \text{ MPa}$ $S_p = 580 \text{ MPa}$	
Design Inputs	- Total mass m - Gravity g - Overhead factor λ - Number of bolts N - Hole diameter d_h - Plate thickness t	$m = 200 \text{ kg}, g = 9.81, \lambda = 2.5, t = 8 \text{ mm}$
Design Weight	$W_d = \lambda mg$	$2.5 \times 200 \times 9.81 = 4.905 \text{ kN}$
Selected bolts and grades	Based on load & environment: M10 (8.8) for indoor use, or M12 (A2-70) for corrosion resistance	
Bolt Capacity (Single Shear)	$V_{cap} = 0.6 F_u A_t$ Shear capacity estimated using from the Bolt Failure in Shear concept (MMMB215 p. 54), consistent with	M10 A2-70 = 24.36 kN M10 8.8 = 27.84 kN M12 A2-70 = 35.41 kN M12 8.8 = 40.46 kN

	Shigley's 60 % tensile-to-shear strength relation. [11]	
Minimum bolts for FOS = 3	$N_{min} = \left(\frac{W_d}{\frac{V_{cap}}{3}} \right)$	M10 A2-70 = 1; M10 8.8 = 1; M12 A2-70 = 1; M12 8.8 = 1 → <i>minimum 2 bolts for stability</i>
Final Bolt count	N	N = 4
Shear Per Bolt	$P_i = \frac{W_d}{N}$	4.905/4 = 1.225 kN
Shear FOS	$FOS_{shear} = \frac{V_{cap}}{P_i}$	M10 A2-70 = 19.9 M10 8.8 = 22.7 M12 A2-70 = 28.9 M12 8.8 = 33.0
Plate bearing stress	$\sigma_b = \frac{P_i}{t \cdot d_h}$	M10 = 13.9 MPa; M12 = 11.8 MPa; M16 = 8.5 MPa << 150 MPa
Preload (assembly)	$F_i = 0.75 S_p A_t$	M10 8.8 = 25.2 kN; M12 A2-70 = 28.5 kN
Tightening torque	$T = K F_i d$	M10 8.8 ≈ 50 N·m; M12 A2-70 ≈ 68 N·m

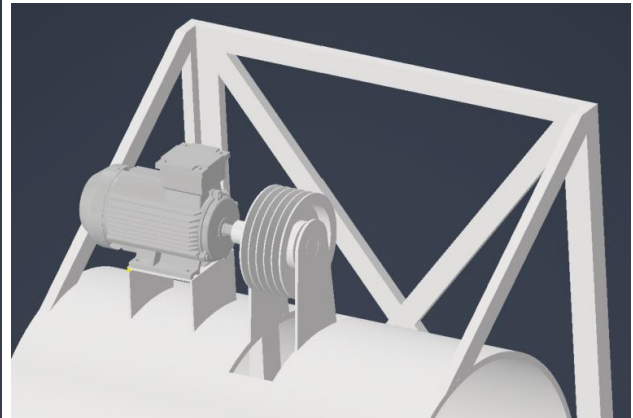
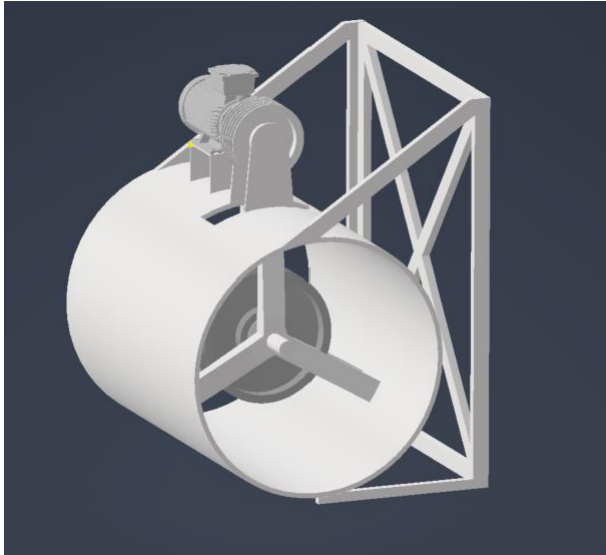
From the above results it can be deduced that:

M10 (8.8) and M12 (A2-70) were selected based on capacity, availability, and environment. M10 provides strength with a high safety margin whilst keeping hole size, plate stress, and installation torque reasonable. Though larger bolts are an option, it will add unnecessary weight, cost, and drilling complexity, whilst smaller bolts on the other hand (M8) have lower stiffness and fatigue resistance for overhead mounting. Class 8.8 offers high yield strength (~640 MPa) for indoor use, while A2-70 stainless offers corrosion protection for outdoor or humid conditions. Both satisfy $\geq 3 \times$ safety factor and maintain practical installation torque levels (≤ 70 N·m). [12]

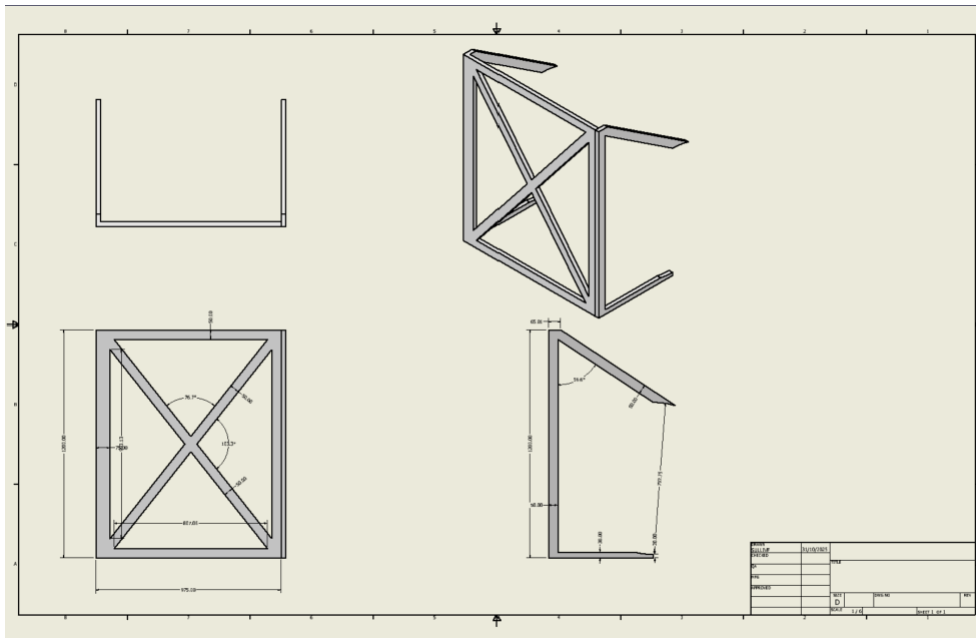
- **Minimum required bolts:** 2 × M10 (8.8) OR M12 (A2-70) (FOS ≥ 3).
- Selected for safety and redundancy: **4 × M10 (8.8)** or **4 × M12 (A2-70)**, both giving **FOS $\gg 3$** under the 200 kg load.
- Plate check: the **8 mm** thick steel plate easily resists bearing stress (≤ 14 MPa << **150 MPa** allowable).
- Assembly torque: **M10 (8.8) ≈ 50 N·m, M12 (A2-70) ≈ 68 N·m.**
- **Final recommendation:**
 - **Indoor / low corrosion:** 4 x M10 × 1.5 ISO 8.8 Hex Flange Bolt, 40 mm long, with Nyloc Nut.
 - **Outdoor / corrosive:** 4 x M12 × 1.75 ISO A2-70 Hex Flange Bolt, 45 mm long, with Nyloc Nuts.

Task 6: CAD Assembly and Drawings

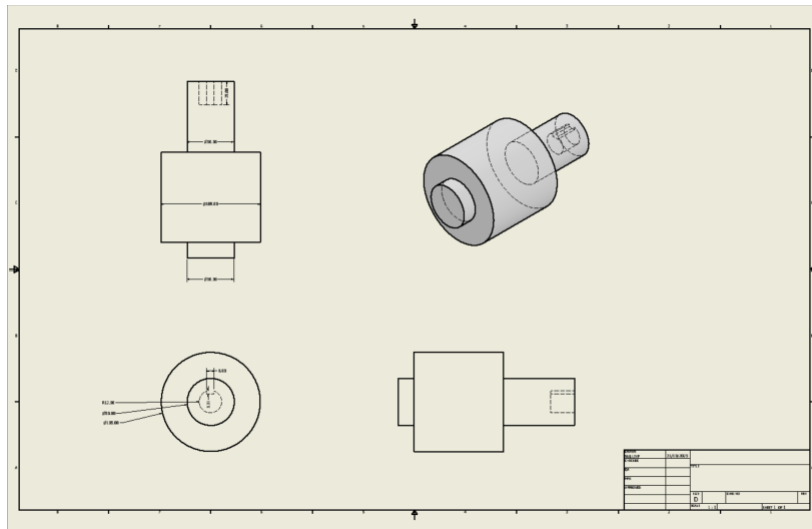
Overall



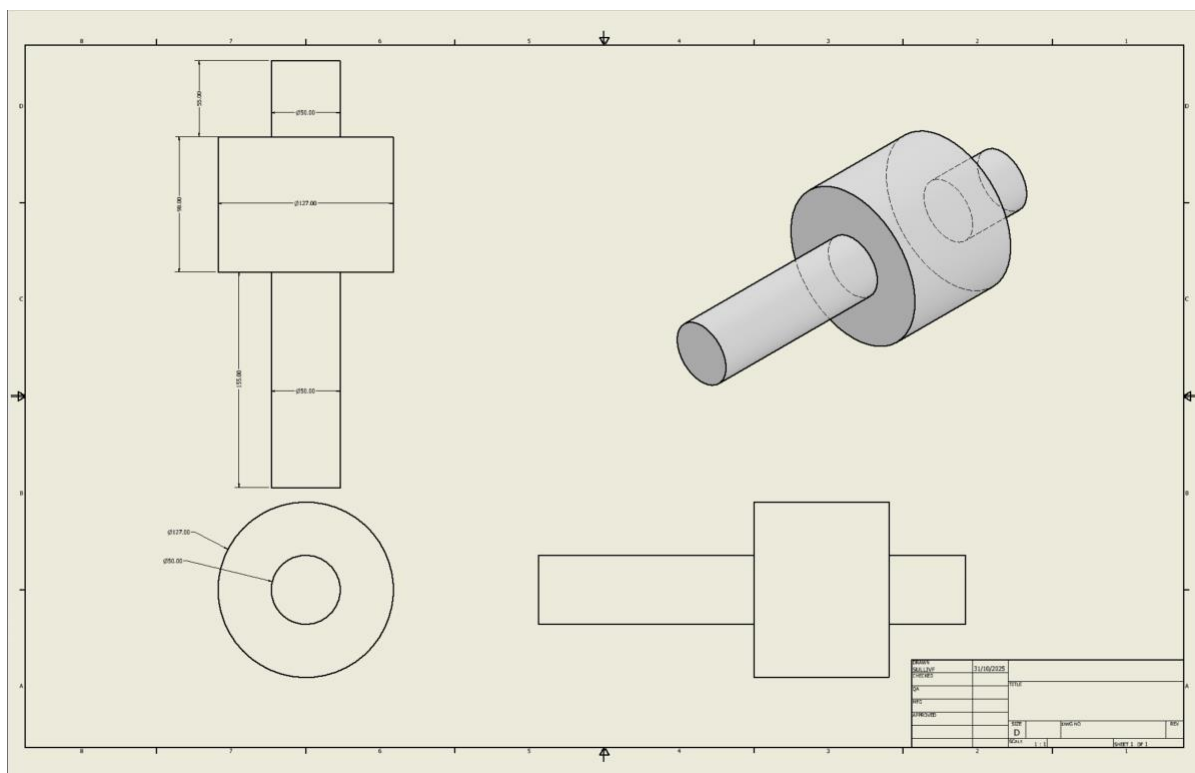
Support Bracket



Shaft 1 (Top)



Shaft 2 (internal)



References

- [1] MAXe3 Low Voltage 3-Phase Induction Motors RANGE 0.55 k W TO 500 k W Totally enclosed fan-cooled cast iron frame series, TECO, (2024), (<https://electricmotors.teco.com.au/wp-content/uploads/sites/53/2025/02/2024-TECO-MAXe3-Catalogue-def-1.pdf>)
- [2] TECO MAX-Ex Premium Efficiency Flameproof Motors RANGE 0.75 k W TO 185 k W Squirrel Cage Induction Motors, TECO, (2025), (<https://electricmotors.teco.com.au/wp-content/uploads/sites/53/2025/09/TECO-MAX-Ex-Premium-Efficiency-Flameproof-Motors-Brochure092025.pdf>)
- [3] TECO MAX-Ex Squirrel Cage Induction Motors RANGE 0.75 k W TO 110 k W Flameproof motors with Increased Safety Terminal Box, TECO, (2025), (<https://electricmotors.teco.com.au/wp-content/uploads/sites/53/2025/03/TECO-MAX-Ex-Exde-Brochure.pdf>)
- [4] TECO MAX-Ex Premium Efficiency Hazardous Area Motors RANGE 0.37 k W TO 450 k W Squirrel Cage Induction Motors, TECO, (2025), (<https://electricmotors.teco.com.au/wp-content/uploads/sites/53/2025/10/TECO-MAX-Ex-Premium-Efficiency-Hazardous-Area-Motors-Brochure.pdf>)
- [5] ASHB & ASFB Quality with Real Economy British and Australian Standards 415/440 volt 3 phase 50Hz Frame 160 To 315 Open Dripproof Enclosure, TECO, (2024), (<https://electricmotors.teco.com.au/wp-content/uploads/sites/53/2024/04/ASFB-ASHB-20140331.pdf>)
- [6] TOTALLY ENCLOSED FAN-COOLED, LOW VOLTAGE SQUIRREL CAGE HIGH THRUST HOLLOWSHAFT PUMP MOTORS, TECO, (2024), (<https://electricmotors.teco.com.au/wp-content/uploads/sites/53/2024/04/AEBH-Catalogue.pdf>)
- [7] TECO Australia and New Zealand Te3 Low Voltage 3-Phase Induction Motors RANGE 0.55 k W TO 500 k W Totally enclosed fan-cooled cast iron frame series, TECO, (2024), (<https://electricmotors.teco.com.au/wp-content/uploads/sites/53/2024/08/TECO-Te3-Brochure-2024.pdf>)
- [8] Belt Drives & Types of Belts, Fractory, Siim Sild, (2022), (<https://fractory.com/belt-drives/>)
- [9] Timing Belt, V-Belt and Flat Belt Design and Engineering Formulas, Engineers Edge, (2025), (https://www.engineersedge.com/vee_flat_belt_menu.shtml)
- [10] V-Belts and Flats Belts: Learn Their Types, Advantages & Differences, Testbook, (2023), (<https://blogmedia.testbook.com/blog/wp-content/uploads/2023/03/v-belts-and-flat-belts-explained-with-types-pros-and-cons-dd622c4b.pdf>)
- [11] University of Wollongong (2025) *MMMB215 Screws and Fasteners (Lecture)* [Lecture notes]. *MMMB215 Mechanical Design 1*, University of Wollongong, 26 October 2025.

[12] B&B Halmstad (2025) *Fastener class (ISO 898-1)*. Available at:
<https://www.bbshalmstad.se/en/info-center/fastener-class-iso-898-1/> (Accessed: 26 October 2025)